

Stronger multiplicity one theorems for forms of general type on GL_2

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In honour of Professor Heini Halberstam

1. Introduction

Let π_1, π_2 be cuspidal automorphic representations of $GL_2(\mathbf{A}_{\mathbf{Q}})$. Let S be a fixed finite set of finite places of $\mathbf{Q}$. The strong multiplicity one theorem for GL_2 states that if $\pi_{1,v} \simeq \pi_{2,v}$, for all places v of $\mathbf{Q}$ not in S , then the representations π_1 and π_2 coincide, where $\pi_{i,v}$, $i = 1, 2$ denotes the local components of π_i . This is equivalent to saying that the Fourier coefficients satisfy $a_p(\pi_1) = a_p(\pi_2)$, for all primes p not belonging to the set S .

Let us now consider the Rankin-Selberg convolutions of the symmetric powers $\text{Sym}^m(\pi_1) \times \text{Sym}^n(\pi_2)$ attached to the automorphic representations π_1 and π_2 and let $L_{m,n}(s)$ be their associated L -series. Each of these series admits an Euler product, which is absolutely convergent in some half plane. Suppose now that each of these L -functions $L_{m,n}$ have an analytic continuation to the entire complex plane, are entire, have a functional equation and are nonvanishing at the edge of the region of absolute convergence. Assume further that GRH holds for $L_{m,n}$. For x a positive real number, let

$$C(\pi_1, \pi_2, x) := \#\{p \leq x \mid a_p(\pi_1) = a_p(\pi_2)\}$$

Our main result states that under our hypothesis above

$$C(\pi_1, \pi_2, x) = O(x^{\frac{5}{6} + \epsilon})$$

In particular the exceptional set S of primes for which the Fourier coefficients do not match, is of Dirichlet density one. In ([11]), D. Ramakrishnan showed unconditionally that if the Fourier coefficients of two cuspidal automorphic representations on GL_2 agree outside a set of primes having Dirichlet density at most $1/8$, then the representations coincide.

Our results can also be interpreted as a 'variable' case of the conjectures of Lang-Trotter. Let E be an elliptic curve without CM and let

$$\pi_{E,a}(x) = \#\{p \leq x \mid a_p(E) = a\}$$

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where for a prime p of good reduction for E , $p + 1 - a_p(E)$ is the number of points of $E(\text{mod } p)$. Lang-Trotter ([8]) conjecture that

$$\pi_E(x) \asymp C_E \frac{x^{\frac{1}{2}}}{\log x}$$

Let ρ_l be the associated system of l -adic representations of E . It has been shown by J. P. Serre ([13]) under the hypothesis of GRH for certain Artin L -functions that

$$\pi_{E,a}(x) = \begin{cases} O(x^{\frac{3}{4}}) & a = 0 \\ O(x^{\frac{5}{6}}/(\log x)^{\frac{1}{3}}) & a \neq \pm 2 \end{cases}$$

Under GRH for L -functions attached to $\text{Sym}^n(\rho_l)$ for all n , it was shown by V. K. Murty ([10]) that

$$\pi_{E,a}(x) = O(x^{\frac{3}{4}}(\log Nx)^{\frac{1}{2}}) \quad \forall a$$

where N is the conductor of E . Our methods also allow us to obtain this result under the same hypothesis as in ([10]).

In ([2]), N. Elkies has shown that there are infinitely many supersingular primes for any elliptic curve E over $\mathbf{Q}$. The distribution of supersingular primes has been studied by N. Elkies and M. R. Murty ([3, pages 25,26]; [4]), who showed unconditionally that

$$\pi_{E,0}(x) = O(x^{\frac{3}{4}})$$

E. Fouvry and M. R. Murty ([5]) established a lower bound for the number of supersingular primes. For any positive δ , they showed that for sufficiently large enough x depending only on E and δ , the following lower bound:

$$\pi_{E,0}(x) \geq \frac{\log_3 x}{(\log_4 x)^{1+\delta}}$$

where $\log_k x$ denotes the k -fold iterated logarithm function. They also established a Ω -type bound, i.e., there exists a constant $c > 0$, such that

$$\pi_{E,0}(x) \geq c \log_2 x$$

for infinitely many $x \rightarrow \infty$. Moreover they have shown that the Lang-Trotter conjecture to be true on 'the average'. In ([6]), they also consider the distribution of the supersingular primes common to two elliptic curves without CM and show that on the average the behavior is like $\log \log x$.

Our proof is similar to ([10]). However we use the notion of discrepancy of a sequence. We invoke a result of Erdős-Turán ([9]), which provides an

upper bound for the discrepancy of a sequence of real numbers in terms of certain exponential sums. We then make use of the explicit formula method to estimate these exponential sums.

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2. Preliminaries

We will work throughout with holomorphic cuspidal eigenforms of weight two. The general situation can also be handled in a similar manner. Let π be an automorphic representation corresponding to a normalised newform f of weight two and level N . The associated L -series is of the form

$$L(s, f) = \prod_{p|N} L_p(s, f) \prod_{(p, N)=1} (1 - a_p p^{-s} + p^{1-2s})^{-1}$$

where a_p are the Hecke eigenvalues of f . $L_p(s, f)$ are certain Euler factors to be attached at the ramified primes $p|N$ of f , in order to get a suitable L -function $L(s, f)$, which satisfies a functional equation of the following form. Let $\Gamma_{\mathbf{C}}(s) = (2\pi)^{-s} \Gamma(s)$ and set $\Lambda(s) = N^{s/2} \Gamma_{\mathbf{C}}(s) L(s, f)$. Let us assume now that our Dirichlet series has real coefficients. Then $\Lambda(s, f)$ satisfies a functional equation of the form

$$\Lambda(s, f) = \pm \Lambda(2 - s, f)$$

We remark that in order to obtain the functional equation in the standard form $s \rightarrow 1 - s$, we have to shift the function $L(s)$ by $s \rightarrow s + 1/2$. Then the shifted L -function is of the form

$$L(s, f) = \prod_{p|N} L_p(s + 1/2, f) \prod_{(p, N)=1} (1 - \alpha_p p^{-s})^{-1} (1 - \bar{\alpha}_p p^{-s})^{-1}$$

where for $(p, N) = 1$, $|\alpha_p| = 1$ and $a_p = (\alpha_p + \bar{\alpha}_p)p^{1/2}$.

The partial L -series naturally associated to $Sym^n(\pi)$ is

$$L_n(s, \pi) := L(s, Sym^n(\pi)) = \prod_{(p, N)=1} \prod_{k=0}^n (1 - \alpha_p^{n-2k} p^{-s})^{-1}$$

Under our hypothesis above this product converges absolutely for $\text{Re}(s) > 1$. To complete this L -function by adding the factors at the ramified primes, we

invoke the local Langlands correspondence. For a prime (archimedean or not) v of $\mathbf{Q}$, let W_v be the local Weil group ([15]) attached to $\mathbf{Q}_v$. By the local Langlands correspondence proved by Kutzko, it is possible to attach to the local component π_v of π , a representation $\sigma_v(\pi_v)$ of the local Weil group W_v into $GL(2, \mathbf{C})$. Then the local L -factor $L_v(s, \text{Sym}^n(\pi))$ is defined as

$$L_v(s, \text{Sym}^n(\pi)) = L(s, \text{Sym}^n(\sigma_v(\pi_v)))$$

where the right hand side is defined as in ([14]). Then the completed L -function, which we continue to denote by $L_n(s, \pi)$ is

$$L_n(s, \pi) = \prod_v L_v(s, \text{Sym}^n(\pi))$$

where the product is over the equivalence classes of primes of $\mathbf{Q}$.

Let $\Gamma_{\mathbf{R}}(s) = \pi^{-s/2} \Gamma(s/2)$. From the prescription given by ([14]) and by a knowledge of the weights that occur in the n^{th} symmetric power representation of the regular representation of $GL_2(\mathbf{C})$, it is possible to calculate the Γ -factors to be attached to $L_n(s, \pi)$ in order to obtain the (conjectural) functional equation to be satisfied by $L_n(s, \pi)$. For each positive integer n , let

$$\gamma_n(s) = \begin{cases} \Gamma_{\mathbf{R}}(s - n/2) & \text{if } n \equiv 0 \pmod{4} \\ \Gamma_{\mathbf{R}}(s - n/2 + 1) & \text{if } n \equiv 2 \pmod{4} \\ 1 & \text{if } n \text{ is odd} \end{cases}$$

$$\Gamma_n(s) = \Gamma_{\mathbf{C}}(s) \Gamma_{\mathbf{C}}(s-1) \dots \Gamma_{\mathbf{C}}\left(s - \frac{n-1}{2}\right) \gamma_n(s)$$

Let $\Lambda_n(s, \pi) = AN_n^{s/2} \Gamma_n(s + n/2) L(s, \pi)$ for some suitable constants A and N_n . Then it is conjectured that

$$\Lambda_n(s) = \pm \Lambda_n(1-s)$$

Let now π_1, π_2 be cuspidal automorphic representations of $GL_2(\mathbf{A}_{\mathbf{Q}})$ corresponding to forms of weight 2. The partial L -function $L_{m,n}(s)$ attached to $\text{Sym}^m(\pi_1) \times \text{Sym}^n(\pi_2)$ is

$$\begin{aligned} L_{m,n}(s) &:= L(s, \text{Sym}^m(\pi_1) \times \text{Sym}^n(\pi_2)) \\ &= \prod_p \left\{ \prod_{k=0}^m \prod_{l=0}^n (1 - \alpha_p^{m-2k} \beta_p^{n-2l} p^{-s})^{-1} \right\} \end{aligned}$$

where α_p (resp. β_p) are the invariants attached to π_1 (resp. π_2) at the prime p . Let us assume that $m > n$. From the decomposition

$$\mathrm{Sym}^m(\mathbf{C}^2) \otimes \mathrm{Sym}^n(\mathbf{C}^2) \simeq \bigoplus_{i=0}^{2n} \mathrm{Sym}^{m+n-2i}(\mathbf{C}^2)$$

as $GL_2(\mathbf{C})$ -modules, we can calculate similarly the Γ factors to be attached to $L_{m,n}(s)$ in order to obtain the form of the functional equation. Let

$$\Lambda_{m,n}(s) = A_{m,n} N_{m,n}^{s/2} \prod_{i=0}^{2n} \Gamma_{m+n-2i}(s + m/2 + n/2) L_{m,n}(s)$$

Then it is conjectured that

$$\Lambda_{m,n}(s) = \pm \Lambda_{m,n}(1-s)$$

We also need to estimate the conductors. It follows from the prescription given in ([12]), and by the local Langlands correspondence, that in the situation above

$$N_k \leq N^k, N_{m,n} \leq (N_1 N_2)^{nm}$$

3. Explicit formula

The explicit formula method allows one to relate sums of Dirichlet coefficients of suitable L -functions, to a sum over the zeros of the L -function in the critical strip. We recall the essential features of the method. For further details about the explicit formula method, we refer to ([1], [7], [10]).

Let K be a number field of degree n_K over $\mathbf{Q}$. Let v denote an arbitrary place of K . Let $L(s)$ be a Dirichlet series which converges absolutely in the half-plane $\mathrm{Re}(s) > 1$. We will assume that $L(s)$ satisfies the following properties EF1-EF4.

EF1 (*Euler product*): $L(s)$ has an Euler product of the form

$$L(s) = \prod_v P_v((Nv)^{-s})^{-1}$$

in the region $\mathrm{Re}(s) > 1$, where v runs over all the finite places of K . Further there exists a finite set S of finite places of K , such that P_v is assumed to be a polynomial of degree $\leq d$, and for $v \notin S$, the degree of P_v is precisely d .

EF2 (*Deligne-Ramanujan estimates*): $P_v(T)$ has the following form:

$$P_v(T) = \prod_i (1 - w_{i,v} T)$$

where the reciprocal roots satisfy $|w_{i,v}| \leq 1$, and $|w_{i,v}| = 1$ for $v \notin S$.

EF3 (*Functional Equation*): Let

$$\check{L}(s) = \prod_v \prod_i (1 - \bar{w}_{i,v} N v^{-s})^{-1}$$

Let λ run over a finite indexing set, the indexing set depending on the archimedean components, and let $s_\lambda = s + i\phi_\lambda$, $\check{s}_\lambda = s - i\phi_\lambda$ for some real number ϕ_λ . Define

$$L_\infty(s) = \prod_\lambda \Gamma\left(\frac{n_\lambda s_\lambda + f_\lambda}{2}\right), \quad \check{L}_\infty(s) = \prod_\lambda \Gamma\left(\frac{n_\lambda \check{s}_\lambda + f_\lambda}{2}\right)$$

where $n_\lambda = 1$ or 2 , f_λ are non-negative integers. Let $p_\lambda = |\phi_\lambda| + f_\lambda$. Set

$$\Lambda(s) = AN^{s/2} L_\infty(s) L(s), \quad \check{\Lambda}(s) = AN^{s/2} \check{L}_\infty(s) \check{L}(s)$$

where N is some suitable constant depending on L , the conductor of L . A is a constant depending again on L , which will not play a role in what we are going to do. We assume that $\Lambda(s)$ and $\check{\Lambda}(s)$ have an analytic continuation to the entire complex plane, are holomorphic everywhere except for possible poles at $s = 0$ and 1 , and that they satisfy a functional equation of the form

$$\Lambda(s) = w \check{\Lambda}(1-s)$$

with $|w| = 1$.

EF4 (*Hadamard factorisation*): Let the order of pole of $\Lambda(s)$ at $s = 0$ (or $s = 1$ by the functional equation) be r . We assume that $(s(s-1))^r \Lambda(s)$ is an entire function of order 1, and so there is a Hadamard product decomposition of the form

$$\Lambda(s) = [s(s-1)]^{-r} a e^{bs} \prod_\rho (1 - s/\rho) e^{s/\rho}$$

where ρ runs over the zeros of $\Lambda(s)$.

Let $\Omega(v, n) = \sum_{i=1}^d w_{i,v}^n$. Then

$$\frac{L'}{L}(s) = \sum_{v,n} \Omega(v, n) \log(Nv) (Nv)^{-ns}$$

where Nv is the norm of v .

The first step in the explicit formula method is to use the discontinuous integral to obtain

$$\sum_{(Nv)^n < x} \Omega(v, n) \log(Nv) = -\frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{L'}{L}(s) x^s \frac{ds}{s} + R_1(x, T, K, L)$$

for $c = 1 + 1/\log(x)$ with

$$R_1(x, T, K, L) \ll n_K dx (\log x)^2 / T$$

where by the Vinogradov notation $a \ll b$, we mean that $|a| \leq C|b|$ for some absolute constant C . Next we shift the line of integration leftwards. We have

$$-\frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{L'}{L}(s) x^s \frac{ds}{s} = -(I_1 + \dots + I_5) + S$$

where $I_1, \dots, I_5$ are integrals along the following segments:

$$I_1 \text{ is on } -(1/4) < \sigma \leq c, t = -T$$

$$I_2 \text{ is on } -U \leq \sigma \leq -1/4, t = -T$$

$$I_3 \text{ is on } \sigma = -U, |t| \leq T$$

$$I_4 \text{ is on } -U \leq \sigma \leq -1/4, t = T$$

$$I_5 \text{ is on } -1/4 < \sigma \leq c, t = T$$

where $U = 1/4 + j$, j is a large positive integer and T is chosen so that $L(\sigma \pm iT) \neq 0$ for any σ . S denotes the sum of the residues at poles of the integrand inside this contour and will be the 'main' term.

I_2, I_3 and I_4 are estimated using the functional equation and estimates on the Γ -function, the 'fudge' factors of the functional equation. To estimate I_1 and I_5 , we also use the assumption that $\Lambda(s)$ is an entire function of order 1. Let $N(t)$ denote the number of zeros $\rho = \beta + i\gamma$, $0 < \beta < 1$, $|\gamma - t| \leq 1$ of $L(s)$. The crucial proposition we need is to control the growth of $N(t)$. From the functional equation and the existence of the Hadamard factorisation it can be deduced along the lines given in ([10]), that

$$N(t) \ll \log N + dn_K + \sum_{\lambda} \log(|t| + p_{\lambda} + 4)$$

where $p_{\lambda} = |\phi_{\lambda}| + f_{\lambda}$. We have

$$-\frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{L'}{L}(s) x^s \frac{ds}{s} = S + R_2(x, T, U, L, K)$$

and R_2 satisfies the following estimate:

$$\begin{aligned} R_2 &\ll \frac{x \log x}{T} \log \{N e^{dn_K} \Pi_\lambda(n_\lambda(T + p_\lambda + 5))\} \\ &+ \frac{x^{-U}}{U} T \log \{N e^{dn_K} \Pi_\lambda(n_\lambda(T + U + p_\lambda + 5))\} \\ &+ \frac{xr}{T^2} + \frac{xr}{T \log x} \end{aligned}$$

The contributions to S are as follows:

- a) A zero or pole of order r of $L(s)$ at $s = 1$. This contributes a term rx .
- b) Poles coming from the zeroes of $L(s)$ in $0 < \sigma < 1$, which contribute a sum

$$- \sum_{|\operatorname{Im} \rho| < T} \frac{x^\rho}{\rho}$$

- c) Poles coming from the trivial zeroes, which are the zeroes of the Gamma factors. From the functional equation, it follows that for $x > 1$, this is bounded by the sum

$$\sum_{\lambda} n_{\lambda} \sum_m \frac{x^{-(2m+f_{\lambda})/n_{\lambda}}}{2m+f_{\lambda}} \ll n_K/x$$

- d) a possible double pole at $s = 0$. Let α be the number of λ such that p_{λ} is zero. Then the contribution of this term is of $(\alpha - r) \log x + c$, where the constant c can be explicitly calculated. Here α is the number of λ such that $\phi_{\lambda} = 0$ and $f_{\lambda} = 0$.

$$\begin{aligned} c = & - \sum_{|\rho| < 1/2} \frac{1}{\rho} - \sum_{p_{\lambda} \neq 0} \frac{1}{2} n_{\lambda} \frac{\Gamma'}{\Gamma} \left(\frac{in_{\lambda} \phi_{\lambda} + f_{\lambda}}{2} \right) + O(dn_K) + \frac{11r}{6} \\ & + O(\log(N e^{dn_K+r} \Pi_{\lambda}(4 + n_{\lambda}(1 + |\phi_{\lambda}|) + f_{\lambda}))) \end{aligned}$$

Thus we obtain an expression for sums of the coefficients of L of the form

$$\sum_{(Nv)^n < x} \Omega(v, n) \log(Nv) = S_L(U, T) + R_L(U, T)$$

where S is a sum over the residues and R is an error term. We now let $U \rightarrow \infty$. From the description of S , it can be seen that $S(U, T)$ converges. Choosing T to be $x - 1$, we find

$$R \ll \log(x) \log(N e^{dn_K} \Pi_{\lambda}(n_{\lambda}x + p_{\lambda} + 5))$$

Using the Deligne-Ramanujan estimates, we have

$$\sum_{(Nv)^n < x, n \geq 2} \Omega(v, n) \log(Nv) \ll dn_K x^{1/2}$$

Putting all these estimates together, we get then

$$\begin{aligned} \sum_{Nv < x} \Omega(v, n) \log(Nv) &\ll rx + c - \sum_{|\gamma| \leq T} \frac{x^\rho}{\rho} + O(dn_K x^{1/2}) \\ &\quad + O(\log\{N \prod_\lambda (n_\lambda x + p_\lambda + 5)\}) \end{aligned}$$

We will now assume further that $L(s)$ satisfy the following:

Generalised Riemann Hypothesis. *There are no zeros of $L(s)$ in the region $\operatorname{Re}(s) > \frac{1}{2}$.*

Under this assumption, using the estimate on the growth $N(t)$ of the number of zeros of $L(s)$ in the critical strip, the sum over the zeros in the critical strip can be estimated to be

$$\sum_{|\rho| < T} \frac{x^\rho}{\rho} + \sum_{|\rho| < 1/2} \frac{1}{\rho} \ll x^{1/2} \log x \log \left(N \prod_\lambda (x + p_\lambda + 4) \right)$$

Let $c(\lambda)$ be the number of Γ factors occurring in $\Lambda(s)$ and let $p = \sup_\lambda \{p_\lambda + 4\}$. We thus obtain on the assumption of *GRH* for $L(s)$, that

$$\sum_{Nv < x} \Omega(v, n) \log(Nv) \ll x^{\frac{1}{2}} \log x \log \left(N(x + p_\lambda)^{c(\lambda)} \right)$$

Finally by appealing to partial summation we obtain

Theorem 1. *Assume that $L(s)$ is an entire function possessing the properties EF1-EF4. Under the further assumption that $L(s)$ satisfies GRH, we have*

$$\sum_{Nv < x} \Omega(v, n) \ll x^{\frac{1}{2}} \log \left(N(x + p_\lambda)^{c(\lambda)} \right)$$

4. Discrepancy

The notion of *discrepancy* ([9]), is meant to measure the deviation of a sequence of real numbers from an 'ideal' distribution, for example the uniform distribution. Let $\{x_k\}_{k=1}^n$ be a finite sequence of real numbers. For $0 \leq \alpha < \beta \leq 1$, let

$A([\alpha, \beta) : n)$ be the counting function $\#\{i \leq n | x_i \in [\alpha, \beta) \pmod{1}\}$. Define the discrepancy $D_n = D_n(x_1, \dots, x_n)$ of the sequence as

$$D_n = \sup_{0 \leq \alpha < \beta \leq 1} \left| \frac{A([\alpha, \beta) : n)}{n} - (\beta - \alpha) \right|$$

If $\omega = (x_i)_{i \geq 1}$ is an infinite sequence of real numbers, the discrepancy $D_n(\omega)$ is defined to be the discrepancy of the first n terms of the sequence ω , i.e., $D_n(\omega) := D_n(x_1, \dots, x_n)$. By a theorem of Weyl, ω is uniformly distributed modulo 1, if and only if $\lim_{n \rightarrow \infty} D_n(\omega) = 0$.

For us the importance of the notion of discrepancy is a result of Erdős-Turán which gives an upper bound for the discrepancy of a sequence in terms of exponential sums. The exponential sums can then be estimated by means of the explicit formula method.

Theorem 2. (Erdős-Turán) ([9, p. 112]): For any finite sequence $(x_k)_{k=1}^n$ of real numbers and for any positive integer M , we have

$$D_n \leq \frac{6}{M+1} + \frac{4}{\pi} \sum_{h=1}^M \left(\frac{1}{h} - \frac{1}{M+1} \right) \left| \frac{1}{n} \sum_{k=1}^n e^{2\pi i h x_k} \right|$$

We will now apply the estimates from the explicit formula method to the L -functions $L_{m,n}$ to show that a strong form of the multiplicity one theorem holds under the hypothesis of *GRH* for the L -functions $L_{m,n}$. The precise theorem is:

Theorem 3. Let π_1, π_2 be cuspidal automorphic representations of $GL_2(\mathbf{A}_{\mathbf{Q}})$. Assume that the L -functions $L_{n,n}$ and $L_{n,n-2}$ are entire and satisfy *GRH*. Then

$$\#\{p \leq x | a_p(\pi_1) = a_p(\pi_2)\} \ll x^{\frac{5}{6}} \frac{(\log Nx)^{\frac{1}{3}}}{(\log x)^{\frac{2}{3}}}$$

where $a_p(\pi_1)$ (resp. $a_p(\pi_2)$) are the Fourier coefficients of π_1 (resp. π_2).

Proof. We will assume for simplicity that the automorphic representations correspond to forms of weight two. Let α_p (resp. β_p) be the invariants attached to π_1 (resp. π_2). Under the hypothesis of the theorem the explicit formula method can be applied to $L_{n,n}$ to obtain

$$\left| \sum_{p \leq x} \sum_{k,l=0}^n \alpha_p^{n-2k} \beta_p^{n-2l} \right| \ll n^2 x^{1/2} \log Nx$$

uniformly in n , where we have used the estimate for the conductor of $L_{n,n}$ in terms of the conductors of π_1 and π_2 . Applying again the explicit formula method to $L_{n,n-2}$, we obtain

$$\left| \sum_{p \leq x} \sum_{k=0}^n \sum_{l=0}^{n-2} \alpha_p^{n-2k} \beta_p^{n-2l} \right| \ll n^2 x^{1/2} \log Nx$$

and we obtain a similar estimate on applying the explicit formula method to $L_{n-2,n}$. Combining these estimates together we obtain

$$\left| \sum_{p \leq x} (\alpha_p^n \beta_p^n + \bar{\alpha}_p^n \beta_p^n + \alpha_p^n \bar{\beta}_p^n + \bar{\alpha}_p^n \bar{\beta}_p^n) \right| \ll n^2 x^{1/2} \log Nx$$

Let $\theta_p(\pi_1) \in [0, 1)$ (resp. $\theta_p(\pi_2)$) denote the angle corresponding to $\alpha_p(\pi_1)$ (resp. $\beta_p(\pi_2)$) i.e., we have $e^{2\pi i \theta_p(\pi_1)} = \alpha_p$, and similarly for β_p . Consider now the sequence x_p of numbers, indexed essentially by the primes p , where

$$x_p = \pm \theta_p(\pi_1) \pm \theta_p(\pi_2)$$

In the notation for the discrepancy formula of the last section we then have $n = 4\pi(x) \sim 4x/\log(x)$. Let δ be a small positive number to be chosen later. Substituting in the discrepancy theorem of Erdős-Turán we have

$$\begin{aligned} C(\pi_1, \pi_2, x) &:= \#\{p \leq x \mid a_p(\pi_1) = a_p(\pi_2)\} \\ &\leq \#\{p \leq x \mid \theta_p(\pi_1) - \theta_p(\pi_2) \in (-\delta, \delta)\} \\ &\leq 2\delta\pi(x) + \frac{6\pi(x)}{M+1} + \frac{4\pi(x)}{\pi} \sum_{h=1}^M \left(\frac{1}{h} - \frac{1}{M+1} \right) \left| \frac{1}{\pi(x)} \sum_{p \leq x} e^{2\pi i h x_p} \right| \end{aligned}$$

Using the estimates from the explicit formula obtained above, we can estimate the exponential sum to obtain

$$C(\pi_1, \pi_2, x) \ll \delta x + \frac{x}{M \log x} + M^2 x^{1/2} \log Nx$$

Since M and δ are arbitrary we take M to be

$$M = \frac{x^{1/6}}{(\log(Nx) \log x)^{1/3}}$$

We obtain then

$$C(\pi_1, \pi_2, x) \ll x^{5/6} \frac{(\log(Nx))^{1/3}}{(\log x)^{2/3}}$$

and that finishes the proof of the theorem.

Corollary 1. *Suppose p is the smallest prime number such that $a_p(\pi_1) \neq a_p(\pi_2)$. Then under the assumption that $L_{n,n}(s)$ and $L_{n,n-2}(s)$ are entire and satisfy GRH for $n \geq 1$, we have*

$$p \ll (\log N)^2 (\log \log N)^3$$

Proof. Suppose that for all primes $p \leq x$, $a_p(\pi_1) = a_p(\pi_2)$. Then

$$\pi(x) \ll \frac{x^{5/6} (\log(Nx))^{1/3}}{(\log x)^{2/3}}$$

and then it can be checked that the above estimate for the smallest p is sufficient to force an upper bound.

Remark. This result should be compared with Theorem 21 of Serre ([13]). By using a method due to Faltings the $\log \log N$ factor can be removed. See Exposé IX by Deligne in Seminaire Szpiro, Asterisque n° 127.

Some remarks concerning the hypothesis: Suppose π_1 and π_2 are unitary, cuspidal automorphic representations on GL_n and GL_m respectively. Then it is expected that $L(s, \pi_1 \times \pi_2)$ is entire, unless $n = m$ and π_1 is isomorphic to the contragredient representation of π_2 , in which case $(s-1)L(s, \pi_1 \times \pi_2)$ is entire. Let π be a unitary, cuspidal automorphic representation of GL_2 . Let us also assume that the Langlands correspondence holds, i.e., corresponding to π , there exists a Langlands parameter, which is a representation of the ‘Langlands group’ into $GL_2(\mathbb{C})$. Let us call π to be “sufficiently general” if this parameter map is surjective in an appropriate sense. For example, this should exclude the cuspidal representations which arise by the process of automorphic induction from an idele class character belonging to a quadratic extension of our base field. Then it is to be expected that the representations $\text{Sym}^n(\pi)$ are cuspidal for all n .

Moreover, let us continue to expect that if π_1 and π_2 are “sufficiently general” unitary, cuspidal, automorphic representations on GL_2 , such that $\text{Sym}^n(\pi_1) \simeq \text{Sym}^n(\pi_2)$ holds, then π_1 is a twist of π_2 by a Dirichlet character of order n . Let us call also two “sufficiently general” unitary, cuspidal automorphic representations on GL_2 to be in “general position” if one is not obtained from the other by a twist by a Dirichlet character.

With this collection of hypothesis and conjectures, it is to be expected then, that if π_1 and π_2 are “sufficiently general” unitary, cuspidal automorphic representations on GL_2 , which are in “general position”, then the L -functions $L_{m,n}(s)$ are entire.

Although the hypothesis of *GRH* is evidently quite strong, the hope is that it will hold for a large class of L -functions arising from automorphic representations. We can then expect that the final conclusion of our theorem to hold for pairs of cuspidal automorphic representations of GL_2 , which are “sufficiently general” and in “general position”.

We can also use the explicit formula method and the notion of discrepancy, to recover the results of Serre ([13]) and V. K. Murty ([10]), on the lacunarity of the Fourier coefficients of modular forms. Let f be a cuspidal, newform of level N . Let

$$C(a, f, x) := \#\{p \leq x \mid a_p(f) = a\}$$

Theorem 4. *Assume that $L_n(s, f)$ are entire and satisfy *GRH* for all $n \geq 1$. Then*

$$C(a, f, x) \ll x^{3/4} (\log Nx)^{1/2}$$

This is an improvement on the result of Serre and is also applicable for any value of a . However our hypothesis are different.

Corollary 2. *Let p be the smallest prime such that $a_p(f) \neq 0$. Then under the assumption of holomorphicity and *GRH* for $L_n(s, f)$ for $n \geq 1$, we have*

$$p \ll (\log N)^2 (\log \log N)^3$$

Remark. Suppose the form is ‘dihedral’ in the sense that it arises by the process of automorphic induction from an idele class character belonging to a quadratic extension K of $\mathbf{Q}$. Then the automorphic representation corresponding to the symmetric square of f is not cuspidal. Thus our hypothesis on f , excludes such forms from our consideration.

We can also apply our methods to the situation of elliptic curves. Let E be an elliptic curve over $\mathbf{Q}$. If p is a prime of good reduction for E , define $a_p(E)$ as

$$a_p(E) = p + 1 - \#E(\mathbf{F}_p)$$

Let ρ_l be the compatible system of l -adic representations attached to E . One can then associate the L -functions $L_n(s, E)$ attached to the symmetric n^{th} powers of $\text{Sym}^n(\rho_l)$. Given two elliptic curves E_1 and E_2 over $\mathbf{Q}$, one can associate in a manner analogous to what we did above for pairs of automorphic representations, the L -functions $L_{m,n}$ associated to the tensor product of the l -adic representations $\text{Sym}^n(\rho_{1,l}) \otimes \text{Sym}^m(\rho_{2,l})$ of the l -adic representations

attached to E_1 and E_2 . Suppose now that the elliptic curves E_1 and E_2 are modular, and E_1 is not a quadratic twist of E_2 . Let us also assume that E_1 and E_2 are not CM curves. Then it is to be expected that the corresponding forms are "sufficiently general" and in "general position" and so our results should hold in this situation. This will in particular imply that if $a_p(E_1) = a_p(E_2)$, for a set of primes of positive density, then $E_1 \simeq E_2$.

We will now state a refinement only for the case when one of the elliptic curves considered has CM. Let E_1 be a CM-elliptic curve over a number field K , where K is the field of CM of E_1 . In this case the analytic properties of the L -function of the elliptic curve are determined by that of a suitable grossencharacter ψ on K . Thus we are in the situation where one of the representations is one dimensional. We can then get a better estimate on the conductors. Let us also assume that the curve E_2 is a non CM-modular elliptic curve over K , i.e., the $L(s, E_2) = L(s, \pi)$ for some cuspidal automorphic representation on $GL_2(\mathbf{A}_K)$. The final result is

Theorem 5. *With the above notation, assume that E_1 is a CM elliptic curve. Suppose that the L -functions $L(s, \text{Sym}^n(\pi)\psi^n)$, $L(s, \text{Sym}^n(\pi)\psi^{n-2})$ and $L(s, \text{Sym}^{n-2}(\pi)\psi^n)$ are entire and satisfy GRH. Then*

$$\#\{Np \leq x \mid a_p(E_1) = a_p(E_2)\} \ll x^{3/4}(\log Nx)^{1/2}$$

where p runs over the prime ideals of K , and N is the product of the conductors of E_1 and E_2 .

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